

External-Space Self-Consistency Law

A speculative theoretical paper on a hypothetical temporal quantum machine

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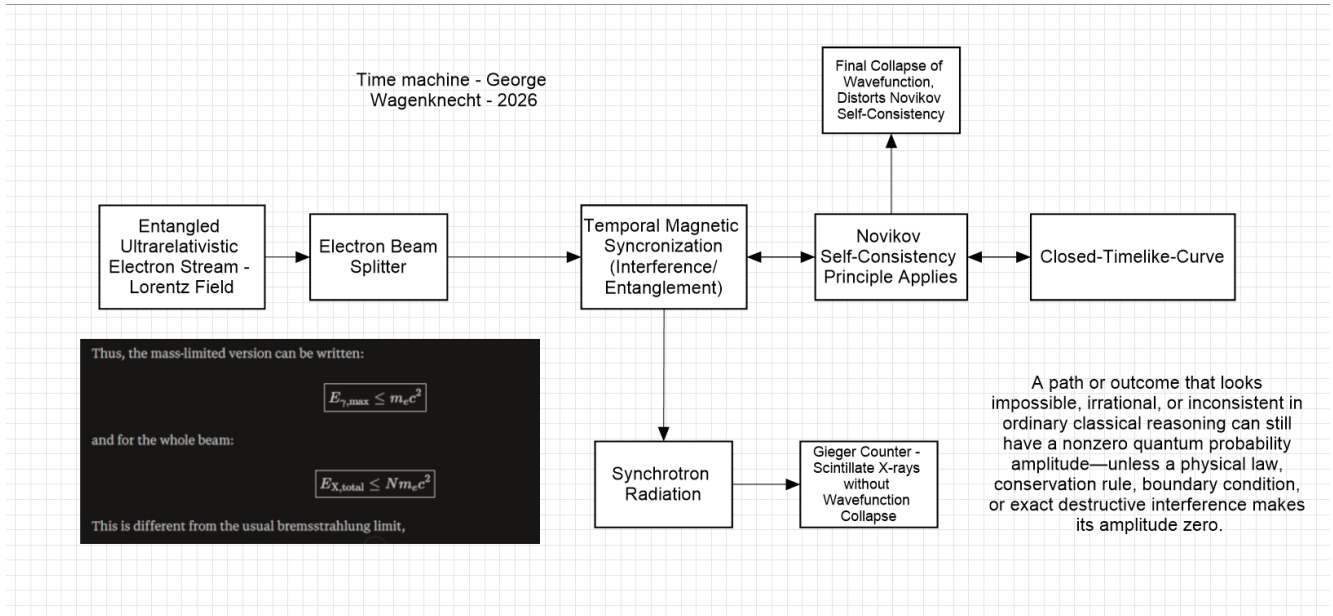


Figure 1. Conceptual architecture of the proposed machine.

Abstract

This paper proposes a speculative principle called the *External-Space Self-Consistency Law*. The principle describes a hypothetical machine in which possible configurations of an external physical space are represented by amplitudes, while a global consistency condition determines which configurations can retain nonzero amplitude. The machine is not assumed to rewrite an already established past. Instead, configurations that cannot satisfy the complete spacetime constraints are suppressed or excluded. This paper develops a mathematical toy model and distinguishes the proposal from experimentally established physics.

1. The Proposed Machine

The machine is treated as a conceptual device rather than a demonstrated time machine. Its architecture is imagined to contain an electron-beam stage, temporal magnetic synchronization, a hypothetical closed-timelike-curve constraint, radiation and a final measurement stage. The defining feature is the proposed global constraint between the machine and the external configuration.

2. External Space as an Amplitude Space

Let the possible externally observable configurations be denoted by $E_{\text{sub-}i}$. A quantum state may be written schematically as:

$$\Psi = \sum \text{over } i \text{ of } A_i \text{ times } E_i$$

Here A_i is the complex amplitude associated with external configuration E_i .

3. External-Space Self-Consistency Law

Law: A physically realizable external configuration must possess a nonzero amplitude under its complete spacetime constraints; configurations that cannot consistently close upon themselves are excluded rather than retroactively altered. Introduce a consistency operator C acting on candidate external configurations.

C acting on $E_i = E_i$, for a self-consistent configuration.

C acting on $E_i = 0$, for an inconsistent configuration.

$\Psi_{\text{allowed}} = C$ acting on Ψ , divided by the square root of the inner product of Ψ with C acting on Ψ .

4. Why the Past Need Not Change

The distinction between rewriting the past and restricting allowed amplitudes is central. A literal retroactive rewrite would transform the state at an earlier time. The proposed law instead says that a complete external configuration inconsistent with the global boundary conditions receives zero or vanishing amplitude. In that interpretation, there is no need for a second past to overwrite the first.

Ψ at past time $\rightarrow \Psi$ -prime at past time

5. Relation to Closed Timelike Curves

The proposal is conceptually related to the Novikov self-consistency principle and to quantum closed-timelike-curve models. Those theories are themselves theoretical frameworks, and different formulations use different mathematical conditions. The present law should therefore be understood as a speculative synthesis-inspired toy model, not an established extension of those theories.

6. Machine-Level Interpretation

Under the proposed model, the electron-beam stage generates candidate alternatives; temporal synchronization defines the relevant boundary condition; the hypothetical consistency region applies the global constraint; radiation provides an observable channel; and the detector samples the resulting external configuration.

7. Global Amplitude Condition

Let $F(E)$ represent the complete temporal, dynamical and boundary constraints on an external configuration E . The proposed law can be expressed compactly as a logical implication: nonzero amplitude requires satisfaction of the global consistency condition. If the constraint fails, the idealized model assigns zero amplitude.

$A(E)$ is nonzero implies $F(E) = 0$

8. Predictions of the Hypothesis

If such a machine existed and the proposed law applied, apparently paradoxical external outcomes would not require the observed past to change. Instead, statistical outcomes could depend on a global consistency condition. A candidate configuration could be mathematically describable before the constraint yet absent from the final observable distribution.

9. Experimental Boundary

No known experiment establishes a macroscopic closed timelike curve or a machine capable of selecting external reality through a temporal consistency operator. The proposed equations therefore describe a conceptual framework, not a demonstrated technology.

10. Conclusion

The central shift is from changing histories to filtering external configurations. Amplitudes represent candidate external states, while a hypothetical global consistency condition determines which candidates survive. The machine therefore need not repair a past event after the fact; the complete spacetime description selects configurations that remain mutually consistent.

Author's note

The External-Space Self-Consistency Law is introduced here as a speculative conceptual law for the proposed machine. It is not presented as an established law of physics. The notation has deliberately been rendered using ordinary text rather than unsupported mathematical glyphs so that the PDF remains legible across viewers.